VELS
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SMART EXPENSE TRACKER AND GOAL BASED SAVING ANALYZER

Abstract

The Smart Expense Tracker and Goal-Based Saving Analyzer was developed using a full-stack approach to provide a secure, user-friendly, and scalable financial management platform. The project began by analysing the problems faced by individuals and groups in tracking shared expenses, managing repayments, and monitoring savings. Based on these requirements, a role-based system architecture was designed with three user levels: Main Administrator, Secondary Administrator, and Normal User. The front end was developed using HTML, CSS, and JavaScript to create a responsive and interactive user interface that works on both desktop and mobile devices. The back end was implemented using Node.js and Express.js, which handle user authentication, expense processing, payment calculations, and administrative operations. User information, expense records, payment history, and saving goals are securely stored in the database using MongoDB/MySQL. The system implements Role-Based Access Control (RBAC) to ensure data privacy. The Main Administrator can create Secondary Administrators and Normal Users, while Secondary Administrators can only create Normal Users. Each Normal User can access only their own financial details, including borrowed amount, repayments, outstanding balance, and savings progress. Additional modules such as expense splitting, payment reminders, budget planning, inventory tracking, visual analytics, report generation, and goal-based saving analysis were integrated to enhance functionality.

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